

# Digital Signal Analysis and Processing Numerical Companion

CT 704 (BEI/BCT, IV-I) / EX 753 (BEX, IV-II) · TU, IOE · tiers over 45 papers, 2066–2082 BS

## Ch 1 • Discrete-Time Signals and Systems

### Numerical Problems & Solutions

**Convolution alone is 34 of the 45 papers.** The four convolution sections below come first for that reason. Within a section the method is worked once, in full, and every other paper that ran the same method is listed under it with its data and its answer: **same process, different numbers, so only the numbers change.**

#### What you need before starting

- $y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$ , and it is commutative, so fold whichever sequence is shorter.
- **Length and position check, use it every time:** for two finite sequences the output starts at  $n_x + n_h$  and has  $L_x + L_h - 1$  samples.
- $\sum_{k=0}^{N-1} a^k = \frac{1-a^N}{1-a}$ ,  $\sum_{k=0}^{\infty} a^k = \frac{1}{1-a}$ ,  $\sum_{k=m}^p a^k = \frac{a^m - a^{p+1}}{1-a}$  ( $a \neq 1$ ).
- $u[n+A] - u[n-B]$  is a **window**: it is 1 for  $-A \leq n \leq B-1$  and 0 elsewhere, so it has  $A+B$  taps. **Getting these two limits right is most of the marks.**

#### 1 Convolution of two finite sequences

**TOP 16/45** 70 Ch **6**, 73 Shr **6**, 70 Bh **2+4**, 70 Asa **2+4**, 74 Bh **6**, 74 Ma **5**, 71 Ch **6**, 79 Bh **5**, 81 Bh **5**, 79 Ba **5**, 74 Ash **3+2**, 82 Bh **5**, 72 Ka **5**, 75 Ash **5**, 79 Ch **4+6**, 75 Bh **4+6** · **one method, sixteen papers**

**QUESTION AS PRINTED** 2070 Chaitra · Q2 · [6]

Find output,  $y(n)$  when:  $h(n) = \{5, 4, 3, 2\}$  and  $x(n) = \{1, 0, 3, 2\}$

#### Method (the tabular method, used for every problem in this section)

1. Write both sequences with their starting index. Here neither is marked, so both start at  $n = 0$ .
2. **Predict the answer's shape first:** start =  $n_x + n_h = 0 + 0 = 0$ , length =  $4 + 4 - 1 = 7$ , so  $y[n]$  runs  $n = 0$  to 6.
3. Build the product table:  $x$  across the top,  $h$  down the side, each cell =  $x[i] h[j]$ .
4.  $y[n]$  is the sum of the cells on the anti-diagonal  $i + j = n$  (shaded steps below).
5. **Check:**  $\sum y[n]$  must equal  $(\sum x[n])(\sum h[n])$ .

#### Step 1: product table

|            | $x[0] = 1$ | $x[1] = 0$ | $x[2] = 3$ | $x[3] = 2$ |
|------------|------------|------------|------------|------------|
| $h[0] = 5$ | 5          | 0          | 15         | 10         |
| $h[1] = 4$ | 4          | 0          | 12         | 8          |
| $h[2] = 3$ | 3          | 0          | 9          | 6          |
| $h[3] = 2$ | 2          | 0          | 6          | 4          |

#### Step 2: sum each anti-diagonal

| $n$ | cells with $i + j = n$ | $y[n]$ |
|-----|------------------------|--------|
| 0   | 5                      | 5      |
| 1   | 4 + 0                  | 4      |
| 2   | 3 + 0 + 15             | 18     |
| 3   | 2 + 0 + 12 + 10        | 24     |
| 4   | 0 + 9 + 8              | 17     |
| 5   | 6 + 6                  | 12     |
| 6   | 4                      | 4      |

## Final answer

$$y[n] = \{\underline{5}, 4, 18, 24, 17, 12, 4\} \quad n = 0 \dots 6$$

The underline marks  $n = 0$ . Check:  $\sum x = 6$ ,  $\sum h = 14$ ,  $6 \times 14 = 84$  and  $5 + 4 + 18 + 24 + 17 + 12 + 4 = 84 \checkmark$

## Every other paper that runs this same method

A window such as  $(1/2)^n \{u[n+2] - u[n-2]\}$  is written out as its samples first: that is the only extra step, and the underlined sample is  $n = 0$ .

| Paper            | Question as printed                                                                                                                                                                                                                                          | $y[n]$                                                                              |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 73 Shr 6         | Determine the system output $y(n)$ of the following signals: $h(n) = \{1, 1, 1\}$ and $x(n) = \{1, 1, 1, 1\}$                                                                                                                                                | $\{\underline{1}, 2, 3, 3, 2, 1\}$                                                  |
| 70 Bh 2+4        | Illustrate the significance of convolution summation in digital signal analysis. Find the output sequence $y(n)$ if: $h(n) = \{1, 1, 1\}$ and $x(n) = \{1-2, 2, 3, 4\}$ the comma after the 1 is missing; it is $\{1, -2, 2, 3, 4\}$                         | $\{\underline{1}, -1, 1, 3, 9, 7, 4\}$                                              |
| 70 Asa 2+4       | Illustrate the significance of convolution summation in digital signal analysis. Compute the convolution of the following signals: $h(n) = \{1, 0, 1\}$ and $x(n) = \{1, -2, -2, 3, 4\}$                                                                     | $\{\underline{1}, -2, -1, 1, 2, 3, 4\}$                                             |
| 74 Bh 6          | Impulse response of an LTI system is given as $h(n) = \{2, 2, -1, 1\}$ . Find the output $y(n)$ of the system to an input $x(n) = \{1, 2, 1, 2\}$ .                                                                                                          | $\{\underline{2}, 6, 5, 5, 5, -1, 2\}$                                              |
| 74 Ma 5          | Find the response of an <b>FIR system</b> with input $x[n] = \{0, 2, 4, 6\}$ and impulse response $h[n] = \{5, 3, 4, 2, 0\}$ .<br><i><math>h</math> and <math>x</math> are named the other way round from every sibling</i>                                  | $\{\underline{0}, 10, 26, 50, 38, 32, 12, 0\}$                                      |
| 71 Ch 6          | A discrete time LTI system has impulse response $h(n) = \{1, 3, 2, -1, 1\}$ <b>for</b> $-1 \leq n \leq 3$ . Determine the system output $y(n)$ if the input $x(n)$ is given by $x(n) = 2\delta(n) - \delta(n-1)$ .                                           | $\{2, \underline{5}, 1, -4, 3, -1\}$ , from $n = -1$                                |
| 79 Bh, 81 Bh 5   | Find the output of LTI system having input signal $x[n] = \delta[n] + 2\delta[n-1] - \delta[n-3]$ and $h[n] = 2\delta[n+1] + 2\delta[n-1]$ .<br>81 Bh prints the same pair, leading with $h$                                                                 | $\{2, \underline{4}, 2, 2, 0, -2\}$ , from $n = -1$                                 |
| 79 Ba 5          | Find the output of LTI system having impulse response $h[n] = (\frac{1}{2})^n \{u[n+2] - u[n-2]\}$ to the input $x[n] = \{2, 1, 0, -1, 4\}$ .                                                                                                                | $\{8, 8, \underline{4}, -2, 14.5, 7, 3.5, 2\}$ , from $n = -2$                      |
| 74 Ash 3+2       | Find the output of LTI system having impulse response $h[n] = (\frac{1}{2})^n \{u[n+2] - u[n-2]\}$ and input signal $x[n] = \{2, 1, 0.5, -1\}$ . <b>Also check the answer.</b><br>same $h$ as 79 Ba, different $x$                                           | $\{8, 8, \underline{6}, -1, -1, -0.75, -0.5\}$ , from $n = -2$                      |
| 82 Bh 5          | Find the output of an LTI system whose impulse response is given by $0.5^n \{u[n] - u[n-3]\}$ and input is given by $\{2, 1, 0.5, -1\}$ .                                                                                                                    | $\{\underline{2}, 2, 1.5, -0.5, -0.375, -0.25\}$                                    |
| 72 Ka 5          | Find the output of LTI system having impulse response $h[n] = (\frac{1}{3})^n \{u[n+1] - u[n-2]\}$ and input signal $x[n] = \{2, 1, 0.5, 3\}$ .                                                                                                              | $\{6, \underline{5}, \frac{19}{6}, \frac{59}{6}, \frac{19}{6}, 1\}$ , from $n = -1$ |
| 75 Ash 5         | Find the output of LTI system having impulse response $h[n] = 2^n \{u[n] - u[n-3]\}$ and input signal $x[n] = \delta[n] + \delta[n-1] + \delta[n-2]$ .                                                                                                       | $\{\underline{1}, 3, 7, 6, 4\}$                                                     |
| 79 Ch, 75 Bh 4+6 | Define the recursive and non-recursive systems with examples. Find the output of LTI system having impulse response $h[n] = 2^n \{u[n] - u[n-3]\}$ and input signal $x[n] = \delta[n] + \delta[n-1] - \delta[n-3]$ .<br>79 Ch prints " <b>2n</b> " for $2^n$ | $\{\underline{1}, 3, 6, 3, -2, -4\}$                                                |

**The two traps in this table.** (1) A  $\delta[n+1]$  or a  $u[n+2]$  puts samples at *negative*  $n$ , so the answer starts left of the origin: mark  $n = 0$  or lose the marks. (2)  $u[n+2] - u[n-2]$  has 4 taps ( $n = -2, -1, 0, 1$ ), not 5: the upper limit  $u[n-B]$  is exclusive.

## 2 Convolution where one sequence is infinite

TOP 12/45 78 Bh 5, 73 Ma 2+6, 78 Ch 6, 76 Ch 6, 69 Ch 1+4, 81 Ch 2+2+6, 80 Ch 2+2+6, 72 Ma 8, 76 Bh 2+6, 77 Ch 3+7, 81 Ba 6, 68 Bh 5, 76 Ash 6, 73 Bh 8 · the answer is a formula per range, not a list of samples

QUESTION AS PRINTED 2078 Bhadra · Q2 · [5]

Find the output of LTI system having impulse response  $h[n] = u[n] - u[n-4]$  and input signal  $x[n] = (1/2)^n u[n]$ .

**Method (used for every problem in this section)**

1. Turn the window into limits.  $h[n] = u[n] - u[n-4]$  is 1 for  $0 \leq n \leq 3$ , so  $h[n-k] = 1$  for  $0 \leq n-k \leq 3$ , i.e.  $n-3 \leq k \leq n$ .
2. Combine with the other sequence's own limits.  $x[k] = (1/2)^k$  needs  $k \geq 0$ . So  $k$  runs from  $\max(0, n-3)$  to  $n$ .
3. The max is why there are two ranges. Solve  $\max(0, n-3)$ :
  - $0 \leq n \leq 3$ : lower limit is 0 (the window hangs off the left of  $x$ ),
  - $n \geq 4$ : lower limit is  $n-3$  (the window is fully inside  $x$ ),
  - $n < 0$ : no overlap at all,  $y[n] = 0$ .
4. Sum the geometric series on each range separately.
5. **Check:** the two formulas must agree in value at the last sample of the first range against a direct count.

**Range A,  $0 \leq n \leq 3$**

$$y[n] = \sum_{k=0}^n \left(\frac{1}{2}\right)^k = \frac{1 - \left(\frac{1}{2}\right)^{n+1}}{1 - \frac{1}{2}} = 2 \left[1 - \left(\frac{1}{2}\right)^{n+1}\right] = \boxed{2 - \left(\frac{1}{2}\right)^n}$$

Sample values:  $y[0] = 1$ ,  $y[1] = 1.5$ ,  $y[2] = 1.75$ ,  $y[3] = 1.875$ .

**Range B,  $n \geq 4$**

$$y[n] = \sum_{k=n-3}^n \left(\frac{1}{2}\right)^k = \frac{\left(\frac{1}{2}\right)^{n-3} - \left(\frac{1}{2}\right)^{n+1}}{1 - \frac{1}{2}} = 2 \left(\frac{1}{2}\right)^{n-3} \left[1 - \left(\frac{1}{2}\right)^4\right] = \boxed{15 \left(\frac{1}{2}\right)^n}$$

Sample values:  $y[4] = 0.9375$ ,  $y[5] = 0.46875$ , and it decays by half each step. **Why the peak is at  $n = 3$ :** that is the last  $n$  at which the whole window still sits over the large early samples of  $x$ .

**Final answer**

$$y[n] = \begin{cases} 0, & n < 0 \\ 2 - \left(\frac{1}{2}\right)^n, & 0 \leq n \leq 3 \\ 15 \left(\frac{1}{2}\right)^n, & n \geq 4 \end{cases}$$

**Every other paper that runs this same method**

Read the table as: window limits  $\rightarrow$  two ranges  $\rightarrow$  two geometric sums. Only the numbers move.

| Paper                                     | Question as printed                                                                                                                                                                                                                                                                                                           | Answer                                                                                                |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 73 Ma 2+6                                 | Define a Causal and Stable system in terms of impulse response. Find the output of LTI system having impulse response $h[n] = u[n+1] - u[n-4]$ and input signal $x[n] = (\frac{1}{2})^n u[n]$ .<br>5 taps, $-1 \dots 3$                                                                                                       | $y = 2 - (\frac{1}{2})^{n+1}$ for $-1 \leq n \leq 3$ ;<br>$y = 31(\frac{1}{2})^{n+1}$ for $n \geq 4$  |
| 78 Ch 6                                   | Find the output of LTI system having impulse response $h[n] = u[n+1] - u[n-3]$ and input signal $x[n] = (0.5)^n u[n]$ .<br>4 taps, $-1 \dots 2$                                                                                                                                                                               | $y = 2 - (\frac{1}{2})^{n+1}$ for $-1 \leq n \leq 2$ ;<br>$y = 15(\frac{1}{2})^{n+1}$ for $n \geq 3$  |
| 76 Ch 6                                   | Find the output of LTI system having input signal $x[n] = u[n+1] - u[n-4]$ and impulse response $h[n] = (\frac{1}{2})^n u[n-1]$ .<br><b>the roles are swapped:</b> here the step is $x$ , not $h$                                                                                                                             | $y = 1 - (\frac{1}{2})^{n+1}$ for $0 \leq n \leq 4$ ;<br>$y = 31(\frac{1}{2})^{n+1}$ for $n \geq 5$   |
| 69 Ch 1+4                                 | Define LTI system. Find the output of LTI system having impulse response $h[n] = 2u[n] - 2u[n-4]$ and input signal $x[n] = (\frac{1}{3})^n u[n]$ .                                                                                                                                                                            | $y = 3[1 - (\frac{1}{3})^{n+1}]$ for $0 \leq n \leq 3$ ;<br>$y = 80(\frac{1}{3})^n$ for $n \geq 4$    |
| 81 Ch 2+2+6 ,<br>80 Ch 2+2+6 ,<br>72 Ma 8 | Define a Causal and Stable system in terms of impulse response. Find the output of LTI system having impulse response $h[n] = 2u[n] - 2u[n-5]$ and input signal $x[n] = (\frac{1}{3})^n u[n]$ .<br>72 Ma prints the design alone, for all 8 marks                                                                             | $y = 3[1 - (\frac{1}{3})^{n+1}]$ for $0 \leq n \leq 4$ ;<br>$y = 242(\frac{1}{3})^n$ for $n \geq 5$   |
| 76 Bh 2+6                                 | Differentiate between FIR and IIR system. Find the output of LTI system having impulse response $h[n] = 2u[n+2] - 2u[n-3]$ and input signal $x[n] = (\frac{1}{3})^n u[n-1]$ .<br>5 taps, $-2 \dots 2$ , and $x$ starts at $n = 1$                                                                                             | $y = 1 - (\frac{1}{3})^{n+2}$ for $-1 \leq n \leq 3$ ;<br>$y = 242(\frac{1}{3})^{n+2}$ for $n \geq 4$ |
| 77 Ch 3+7                                 | Define causality and stability of LTI system in terms of impulse response with a suitable example. Find the output of LTI system having impulse response $h[n] = 2u[n+3] - 2u[n-4]$ and input signal $x[n] = (\frac{1}{2})^n u[n+1]$ .<br>7 taps, $-3 \dots 3$ , and $x$ starts at $n = -1$ : <b>the widest of the family</b> | $y = 8 - (\frac{1}{2})^{n+2}$ for $-4 \leq n \leq 2$ ;<br>$y = 127(\frac{1}{2})^{n+2}$ for $n \geq 3$ |

**Where the 2 in front goes:** an amplitude of 2 on the window (69 Ch, 81 Ch, 76 Bh, 77 Ch) multiplies every term, which is why those constants are 3, 80, 242, 127 rather than 2, 40, 121, 63.5. Multiply at the end, once.

## 2b The four that do not fit the pattern

Each of these changes the calculation, so each is worked separately rather than tabulated.

### 81 Ba 6 : two different exponentials

- Given  $h(n) = (\frac{1}{3})^n u(n-3)$ ,  $x(n) = (\frac{1}{6})^{n-6} u(n)$ .
- First tidy  $x$ :  $(\frac{1}{6})^{n-6} = 6^6 (\frac{1}{6})^n = 46656 (\frac{1}{6})^n$ .
- Overlap needs  $k \geq 0$  and  $n-k \geq 3$ , so  $0 \leq k \leq n-3$ :  
**one range only**,  $n \geq 3$ .

$$y[n] = 46656 \sum_{k=0}^{n-3} \left(\frac{1}{6}\right)^k \left(\frac{1}{3}\right)^{n-k} = 46656 \left(\frac{1}{3}\right)^n \sum_{k=0}^{n-3} \left(\frac{1}{2}\right)^k$$

- The ratio is  $\frac{1/6}{1/3} = \frac{1}{2}$ : **two different bases collapse to one geometric sum**, which is the whole trick here.

$$y[n] = 3456 \left(\frac{1}{3}\right)^{n-3} - 1728 \left(\frac{1}{6}\right)^{n-3}, \quad n \geq 3$$

- Check:  $y[3] = 3456 - 1728 = 1728$ ,  
 $y[4] = 1152 - 288 = 864$ ,  $y[6] = 128 - 8 = 120$ .

### 68 Bh 5 : both sides infinite, one left-sided

- Given  $x[n] = 3^n u[-n-5]$ ,  $h[n] = u[n-5]$ .
- $u[-n-5]$  is 1 for  $n \leq -5$ : **a left-sided signal**, and  $3^n$  decays as  $n \rightarrow -\infty$ , so the sum converges.
- Overlap needs  $k \leq -5$  and  $n-k \geq 5$  i.e.  $k \leq n-5$ .  
Upper limit is  $\min(-5, n-5)$ , lower limit  $-\infty$ .
- $n \leq 0$ : upper limit  $n-5$ , so

$$y[n] = \sum_{k=-\infty}^{n-5} 3^k = \frac{3^{n-5}}{1-\frac{1}{3}} = \boxed{\frac{1}{2} 3^{n-4}}$$

- $n \geq 0$ : upper limit is fixed at  $-5$ , so  $y[n]$  stops changing:

$$y[n] = \sum_{k=-\infty}^{-5} 3^k = \frac{3}{2} \cdot 3^{-5} = \boxed{\frac{1}{162}}$$

- Running the series backwards** is the only new idea: for a left-sided sum put  $k = -m$  and it is an ordinary  $\sum (1/3)^m$ .

**73 Bh 8 : finite × finite, but symbolic**

- Given  $x[n] = 1$  for  $0 \leq n \leq 4$ ,  $h[n] = a^n$  for  $0 \leq n \leq 6$ ,  $a > 1$ .
- Both finite, so the answer runs  $n = 0$  to  $10$ : but  $a$  is symbolic, so it is written as sums, not samples.
- Overlap needs  $0 \leq k \leq 4$  and  $0 \leq n - k \leq 6$ , i.e.  $\max(0, n - 6) \leq k \leq \min(4, n)$ .
- **Two min / max switches means three ranges**, not two.

$$\text{With } S(k) = \frac{a^{k+1} - 1}{a - 1}:$$

$$y[n] = \begin{cases} S(n), & 0 \leq n \leq 4 \\ S(n) - S(n - 5), & 5 \leq n \leq 6 \\ S(6) - S(n - 5), & 7 \leq n \leq 10 \end{cases}$$

- Check with  $a = 2$ :  $y[0] = 1$ ,  $y[4] = 31$ ,  
 $y[6] = 127 - 31 = 96$ ,  $y[10] = 127 - 63 = 64$ .

**76 Ash 6 : the question as printed cannot be solved**

- The paper prints  $x[n] = 2^n 4[-n]$ ,  $0 < a < 1$  and  $h[n] = 4[n]$ : the sequence symbols are mangled, and it is meant to read  $x[n] = a^n u[-n]$ ,  $h[n] = u[n]$ .
- **With  $0 < a < 1$  that diverges.**  $a^n u[-n]$  grows without bound as  $n \rightarrow -\infty$ , so  $\sum_{k \leq \min(0, n)} a^k$  is infinite for every  $n$ .
- **Reading A**, the standard textbook problem and the one that fits the printed  $0 < a < 1$ :  $x[n] = a^n u[n]$ ,  $h[n] = u[n]$ .

$$y[n] = \sum_{k=0}^n a^k = \boxed{\frac{1 - a^{n+1}}{1 - a}} \quad n \geq 0$$

As  $n \rightarrow \infty$  this tends to  $1/(1 - a)$ : a running sum that settles.

- **Reading B**, keeping  $u[-n]$  literally, which needs  $a > 1$  instead:

$$y[n] = \frac{a^{n+1}}{a - 1} \quad (n \leq 0), \quad y[n] = \frac{a}{a - 1} \quad (n \geq 0)$$

- **In the exam:** state the inconsistency in one line, then solve reading A. verified/added

**3 Response to a complex exponential**

**TOP 10/45** 70 Ch 4, 72 Ka 4, 80 Ba 5, 82 Ba 5, 71 Shr 2+3+5, 69 Bh 2+4, 73 Shr 3+7 · **never convolve these** · 5 marks in about 90 seconds

**QUESTION AS PRINTED** 2080 Baishakh · Q2 · [5]

Find the output of LTI system having impulse response  $h[n] = (1/2)^n * u[n]$  and input  $x[n] = 5e^{j\pi n/3}$  for  $-\infty < n < \infty$ .

The paper prints a stray “\*” between  $(1/2)^n$  and  $u[n]$ ; it is a product, not a convolution.

**Method (used for every problem in this section)**

1. Recognise the input: a **single complex exponential running over all  $n$**  is an eigenfunction, so  $y[n] = H(e^{j\omega_0}) x[n]$  and no convolution is needed.
2. Find the frequency response once:

$$H(e^{j\omega}) = \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n e^{-j\omega n} = \sum_{n=0}^{\infty} \left(\frac{1}{2} e^{-j\omega}\right)^n = \frac{1}{1 - \frac{1}{2} e^{-j\omega}}$$

converging because  $|\frac{1}{2} e^{-j\omega}| = \frac{1}{2} < 1$ .

3. Substitute  $\omega_0$  from the input, here  $\omega_0 = \pi/3$ .
4. Put the denominator into rectangular form, then into polar form.
5. Answer = input × that complex number.

**Step-by-step**

- $e^{-j\pi/3} = \cos 60^\circ - j \sin 60^\circ = 0.5 - j0.866$ .
- Denominator =  $1 - \frac{1}{2}(0.5 - j0.866) = 1 - 0.25 + j0.433 = 0.75 + j0.433$ .
- $|0.75 + j0.433| = \sqrt{0.5625 + 0.1875} = \sqrt{0.75} = 0.8660$ ,  $\angle = \tan^{-1} \frac{0.433}{0.75} = 30^\circ$ .
- So  $H(e^{j\pi/3}) = \frac{1}{0.866 \angle 30^\circ} = 1.1547 \angle -30^\circ$ .

$$y[n] = 5 \times 1.1547 e^{-j\pi/6} e^{j\pi n/3} = 5.7735 e^{j(\pi n/3 - \pi/6)}$$

**Read the answer:** the sinusoid keeps its frequency, its amplitude is multiplied by 1.1547 and it is delayed in phase by  $30^\circ$ . That sentence is worth a mark on its own.

**Every other paper that runs this same method**

| Paper                     | Question as printed                                                                                                                                                                                                                                              | $H(e^{j\omega_0})$                                                   | $y[n]$                            |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------|
| 70 Ch 4                   | Find the output of LTI System having impulse response $h[n] = (\frac{1}{3})^n u[n]$ and input signal $x[n] = 5e^{j\pi n/2}$ for $-\infty < n < \infty$ .                                                                                                         | $\frac{1}{1+j/3} = 0.9 - j0.3$<br>$= 0.9487\angle -18.43^\circ$      | $4.7434 e^{j(\pi n/2 - 0.3217)}$  |
| 72 Ka 4, 80 Ba 5, 82 Ba 5 | Find the output of LTI system having impulse response $h[n] = (\frac{1}{2})^n u[n]$ and input $x[n] = 5e^{j\pi n/3}$<br>80 Ba adds "for $-\infty < n < \infty$ "; 82 Ba says "excited by an input"                                                               | $1.1547\angle -30^\circ$                                             | $5.7735 e^{j(\pi n/3 - \pi/6)}$   |
| 69 Bh 2+4                 | Why do you need <b>different equation</b> ? Consider an LTI system with impulse response $h[n] = (\frac{1}{2})^n u[n]$ . Determine $y[n]$ , the output of this system, if the input is $x[n] = Ae^{jn\pi/2}$ "different" for "difference"                        | $\frac{1}{1+j0.5}$<br>$= 0.8944\angle -26.57^\circ$                  | $0.8944A e^{j(\pi n/2 - 0.4636)}$ |
| 71 Shr 2+3+5              | Why do we need difference equation? State linear constant coefficient difference equation and corresponding system function. Consider an LTI system with impulse response $h[n] = (\frac{1}{2})^n u[n]$ . Determine $y[n]$ , if the input is $x[n] = Ae^{jn\pi}$ | $\frac{1}{1+\frac{1}{2}} = \frac{2}{3}$<br>(real, $e^{-j\pi} = -1$ ) | $\frac{2}{3}A e^{j\pi n}$         |

### 73 Shr 3+7 : a sum of three terms, so use superposition

- Given  $h[n] = (\frac{1}{2})^n u[n]$ ,  $x[n] = 10 - 5 \sin(\pi n/2) + 20 \cos(\pi n)$ .
- **Do not convolve.** Split into frequencies and scale each by  $H$  at that frequency.
- $\omega = 0$  (the constant 10):  $H(e^{j0}) = \frac{1}{1-\frac{1}{2}} = 2 \rightarrow$  contributes 20.
- $\omega = \pi$  (the  $20 \cos \pi n$ ):  $H(e^{j\pi}) = \frac{1}{1+\frac{1}{2}} = \frac{2}{3} \rightarrow$  contributes  $\frac{40}{3} \cos(\pi n) = 13.333 \cos(\pi n)$ .
- $\omega = \pi/2$  (the  $-5 \sin$ ):  $H(e^{j\pi/2}) = 0.8944\angle -26.57^\circ \rightarrow$  contributes  $-4.472 \sin(\pi n/2 - 26.57^\circ)$ .

$$y[n] = 20 + 13.333 \cos(\pi n) - 4.472 \sin(\frac{\pi n}{2} - 26.57^\circ)$$

- **Why a real sinusoid keeps its shape:**  $\cos \omega_0 n$  is two exponentials at  $\pm \omega_0$ , and for real  $h[n]$  the two responses are conjugates, so they recombine into one real sinusoid with the magnitude and phase of  $H$ .

## 4 Convolution from tabulated or stem-plot data

**HOT 4/45** 73 Ch 6, 75 Ch 5+2, 72 Ch 3+2, 71 Bh 7 · same arithmetic as §1, **the marks are in the index bookkeeping**

**QUESTION AS PRINTED** 2073 Chaitra · Q2 · [6]

Calculate  $y[n]$ , if  $x[n]$  is  $x[-2] = 0.5$ ,  $x[0] = 1$ ,  $x[1] = 0.75$ ,  $x[3] = 0.5$  and  $n[n]$  is  $n[0] = 1$ ,  $n[1] = 0.75$  and  $n[2] = 0.5$  and verify your result.

The paper writes the impulse response as " $n[n]$ "; it means  $h[n]$ .

### Method (the only difference from §1)

1. **Fill in the gaps with zeros.** The question lists only the non-zero samples, and a missing index is a zero, not an absent sample.
  - $x[n] = \{0.5, 0, \underline{1}, 0.75, 0, 0.5\}$  for  $n = -2 \dots 3$
  - $h[n] = \{\underline{1}, 0.75, 0.5\}$  for  $n = 0 \dots 2$
2. Predict the shape: start  $= -2 + 0 = -2$ , length  $= 6 + 3 - 1 = 8$ , so  $n = -2 \dots 5$ .
3. Product table and anti-diagonals, exactly as in §1.
4. **"Verify your result"** is worth marks: use the sum check.

### Product table

|               | $x[-2] 0.5$ | $x[-1] 0$ | $x[0] 1$ | $x[1] 0.75$ | $x[2] 0$ | $x[3] 0.5$ |
|---------------|-------------|-----------|----------|-------------|----------|------------|
| $h[0] = 1$    | 0.5         | 0         | 1        | 0.75        | 0        | 0.5        |
| $h[1] = 0.75$ | 0.375       | 0         | 0.75     | 0.5625      | 0        | 0.375      |
| $h[2] = 0.5$  | 0.25        | 0         | 0.5      | 0.375       | 0        | 0.25       |

### Anti-diagonal sums

| $n$ | terms              | $y[n]$ |
|-----|--------------------|--------|
| -2  | 0.5                | 0.5    |
| -1  | $0.375 + 0$        | 0.375  |
| 0   | $0.25 + 0 + 1$     | 1.25   |
| 1   | $0 + 0.75 + 0.75$  | 1.5    |
| 2   | $0.5 + 0.5625 + 0$ | 1.0625 |
| 3   | $0.375 + 0 + 0.5$  | 0.875  |
| 4   | $0 + 0.375$        | 0.375  |
| 5   | 0.25               | 0.25   |

Final answer and the verification the question asks for

$$y[n] = \{0.5, 0.375, \underline{1.25}, 1.5, 1.0625, 0.875, 0.375, 0.25\} \quad n = -2 \dots 5$$

$\sum x = 2.75$ ,  $\sum h = 2.25$ , product = 6.1875; and  $0.5 + 0.375 + 1.25 + 1.5 + 1.0625 + 0.875 + 0.375 + 0.25 = 6.1875$  ✓ **This sum check is the standard way to "verify your result"** and takes one line.

Every other paper that runs this same method

| Paper     | Question as printed                                                                                                                                                                                                                                                    | $y[n]$                                                                           |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 75 Ch 5+2 | Find the output of LTI system having impulse response $h[n]$ with $h[-2] = 3$ , $h[0] = 2$ , $h[1] = 1$ and input signal $x[n] = (2)^n$ , for $-1 \leq n \leq 3$ . <b>Also check the answer.</b>                                                                       | $\{1.5, 3, 7, \underline{14.5}, 29, 10, 20, 8\}$ , from $n = -3$                 |
| 72 Ch 3+2 | Find the output of LTI system having impulse response $h[n]$ with $h[-2] = 1$ , $h[0] = 2$ , $h[1] = 3$ and input signal $x[n]$ with $x[0] = 1/2$ , $x[2] = 2$ , $x[3] = 3$ . <b>Also check the answer.</b><br><b>every unlisted sample is zero</b> , including $x[1]$ | $\{0.5, 0, \underline{3}, 4.5, 4, 12, 9\}$ , from $n = -2$                       |
| 71 Bh 7   | Find the output sequence $y[n]$ <b>and verify it</b> for the two stem plots given: $h[0] = 0.5$ , $h[1] = 0.75$ , $h[2] = 1$ ; $x[-2] = 1$ , $x[0] = 0.5$ , $x[2] = 1$ , $x[3] = 0.75$ .<br>both sequences are drawn, not written                                      | $\{0.5, 0.75, \underline{1.25}, 0.375, 1, 1.125, 1.5625, 0.75\}$ , from $n = -2$ |

**75 Ch is the one people get wrong:**  $x[n] = 2^n$  for  $-1 \leq n \leq 3$  starts at  $n = -1$  with the value 0.5, not at  $n = 0$  with the value 1. With  $h$  starting at  $-2$ , the output starts at  $-3$ .

## 5 Energy and power of a given signal

**TOP 10/45** 79 Bh, 69 Ch, 81 Bh, 74 Ash, 79 Ba, 82 Bh, 72 Ka, 68 Bh, 75 Ch, 74 Ma · **almost always the same question as \$6**, asked as one 4-mark part

**QUESTION AS PRINTED** 2079 Bhadra · Q1 · [2+2]

Define energy and power signal. Determine whether the signal  $x[n] = \cos(2\pi n/5) + \sin(\pi n/3)$  is periodic or not. If periodic, find its fundamental period.

**Method (used for every problem in this section)**

- Look at  $|x[n]|$  before computing anything.
  - Bounded and never dying away  $\rightarrow$  power signal,  $E = \infty$ .
  - Dying away or finite length  $\rightarrow$  energy signal,  $P = 0$ .
- Only then evaluate the one quantity that is finite. Computing both wastes time.
- For a **periodic** signal use one period:  $P = \frac{1}{N} \sum_{n=0}^{N-1} |x[n]|^2$ .
- For a **decaying** signal use  $E = \sum_n |x[n]|^2$  and the geometric sum.
- State the verdict as a sentence: " $E = \infty$  and  $P$  is finite and non-zero, so it is a **power** signal."

**Worked: the 79 Bh signal**

- $x[n] = \cos(2\pi n/5) + \sin(\pi n/3)$  is a sum of two sinusoids: bounded, never decays.
- Both parts are periodic (§6 gives  $N = 30$ ), so the sum is periodic  $\rightarrow$  **power signal**,  $E = \infty$ .
- Power without summing: two sinusoids at *different* frequencies are orthogonal over the common period, so their powers add.

$$P = \frac{A_1^2}{2} + \frac{A_2^2}{2} = \frac{1}{2} + \frac{1}{2} = \boxed{1}$$

- **Quotable:** a real sinusoid of amplitude  $A$  has  $P = A^2/2$ ; a complex exponential of amplitude  $A$  has  $P = A^2$  (no factor of 2, because  $|e^{j\theta}| = 1$  rather than an oscillating cosine).

**Every other paper that runs this same method**

**The instruction varies and the marks follow it.** Papers print one of: "Compare between energy signal and power signal. Determine whether the signal ...is energy signal or power signal"; "Define energy and power signal. Check the signal ...is Energy or Power type"; or "Define energy and power signal. Determine whether the signal ...is periodic or non-periodic and if it is periodic, find its fundamental period". **The last one is two questions**, and its second half is §6. The definition is always worth 2 marks on its own.

| Paper                     | $x[n]$                           | Answer                                                                            |
|---------------------------|----------------------------------|-----------------------------------------------------------------------------------|
| 79 Bh, 69 Ch, 81 Bh 2+2 4 | $\cos(2\pi n/5) + \sin(\pi n/3)$ | power signal, $P = 1$ , $E = \infty$ ; periodic, $N = 30$                         |
| 74 Ash 2+2                | $e^{j(\pi n/3 + \pi/4)}$         | $ x[n]  = 1 \rightarrow$ power signal, $P = 1$ , $E = \infty$ ; periodic, $N = 6$ |
| 79 Ba, 82 Bh 2+2          | $e^{j(\pi n/2 + 4\pi/7)}$        | power signal, $P = 1$ , $E = \infty$ ; periodic, $N = 4$                          |
| 72 Ka 2+3                 | $u[n]$ and $\delta[n]$           | $u[n]$ : power, $P = \frac{1}{2}$ . $\delta[n]$ : energy, $E = 1$                 |
| 68 Bh 5                   | $u[n]$                           | $E = \infty$ , $P = \frac{1}{2}$                                                  |

75 Ch 3+4 and 74 Ma 3+3 ask the *definitions* only (power Vs energy, Fourier series Vs Fourier transform; energy Vs power, periodic Vs aperiodic, symmetric Vs antisymmetric), plus "is  $y[n] = x^2[n]$  linear": that last part is §9. **A constant phase offset like  $+\pi/4$  or  $+4\pi/7$  changes nothing:** it shifts the signal, and neither power nor period depends on a shift.

**6 Periodicity and fundamental period**

**TOP 17/45** 80 Bh 5, 80 Ba 2+2, 78 Bh 4, 70 Ch 3, 69 Bh 4, 67 Mng 4, 66 Ma 4, and the 10 papers of §5 · **the most reliable 4 marks in the paper**

QUESTION AS PRINTED 2080 Bhadra · Q1 · [5]

Determine whether the signal  $x[n] = e^{j\pi n/16} \cos(n\pi/17)$  is periodic or not. If periodic, find its fundamental period.

**Method (used for every problem in this section)**

1. Read off  $\omega_0$  for each factor or term.
2. Form  $\frac{\omega_0}{2\pi}$ . **Irrational  $\rightarrow$  aperiodic, stop there.**
3. Rational: write it as  $\frac{m}{N}$  in **lowest terms**;  $N$  is that term's period.
4. Combine: for a sum or a product of periodic terms, the overall period is  $\text{LCM}(N_1, N_2)$ .
5. **Check by substitution** if unsure: confirm  $x[n + N] = x[n]$  and that no smaller divisor of  $N$  works.

**68 Bh 5: "find the energy and power of  $x[n] = u[n]$ "**

- **Energy:**  $E = \sum_{n=-\infty}^{\infty} |u[n]|^2 = \sum_{n=0}^{\infty} 1 = \boxed{\infty}$
- **Power:** the limit runs over  $2N + 1$  samples but only  $N + 1$  of them are non-zero:

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N + 1} \sum_{n=0}^N 1 = \lim_{N \rightarrow \infty} \frac{N + 1}{2N + 1} = \boxed{\frac{1}{2}}$$

- $P = \frac{1}{2}$ , **not 1**. This is the single most common slip in the chapter, and 72 Ka asks the same thing beside  $\delta[n]$ .
- $x[n] = \delta[n]$  (72 Ka):  $E = 1$ , and  $P = \lim_{N \rightarrow \infty} \frac{1}{2N + 1} = 0 \rightarrow$  **energy signal**.

**Worked: the 80 Bh signal**

- First factor  $e^{j\pi n/16}$ :  $\omega_1 = \frac{\pi}{16}$ , so  $\frac{\omega_1}{2\pi} = \frac{1}{32} \rightarrow N_1 = 32$ .
- Second factor  $\cos(n\pi/17)$ :  $\omega_2 = \frac{\pi}{17}$ , so  $\frac{\omega_2}{2\pi} = \frac{1}{34} \rightarrow N_2 = 34$ .
- Both rational, so the product is periodic.
- $32 = 2^5$ ,  $34 = 2 \times 17$ , so  $\text{LCM} = 2^5 \times 17$ .

$$N = 544$$

**Do not multiply  $32 \times 34 = 1088$ .** The LCM is the answer; multiplying is only right when the two are coprime.

**The three traps, each a real paper**

- 70 Ch (i)**  $\cos(\pi n^2/8)$ :  $n^2$ , so it is **not a sinusoid in  $n$**  and the  $\omega_0/2\pi$  rule does not apply. Test directly:

$$\cos\left(\frac{\pi(n+8)^2}{8}\right) = \cos\left(\frac{\pi n^2}{8} + 2\pi n + 8\pi\right) = \cos\left(\frac{\pi n^2}{8}\right)$$

since  $2\pi n$  and  $8\pi$  are whole multiples of  $2\pi$ .  $N = 4$  fails, so  $N = 8$ .

- 70 Ch (ii)**  $\cos(n/2) \cos(\pi n/4)$ : the first factor has  $\omega = \frac{1}{2}$  with **no  $\pi$** , so  $\omega/2\pi = 1/4\pi$  is irrational  $\rightarrow$  the product is **aperiodic**. One irrational factor is enough.
- 80 Ba (a)**  $\sin(n\pi) + \cos(n\pi)$ :  **$\sin(n\pi) = 0$  for every integer  $n$** , so the signal is just  $\cos(n\pi) = (-1)^n$  and  $N = 2$ . Treating it as  $\text{LCM}(2, 2)$  gives the same answer, but say why the sine vanished.

**Every other paper that runs this same method**

**Every one of these prints the same two-part instruction**, in one of these forms: "Determine if the signal ...is periodic or not. If the signal is periodic, find its fundamental period"; "Determine whether the given signal is periodic or not. If the signal is periodic, determine the fundamental period"; "Check whether following signals are periodic or not. If yes, state their periodic time"; or "Determine which of the following signals are periodic and compute their fundamental period". **Answering only "periodic" loses half the marks** every time.

| Paper        | $x[n]$                                            | $\omega_0/2\pi$ and answer                                                       |
|--------------|---------------------------------------------------|----------------------------------------------------------------------------------|
| 78 Bh 4      | $\cos(\pi n/2) \cos(\pi n/4)$                     | $\frac{1}{4}$ and $\frac{1}{8} \rightarrow N_1 = 4, N_2 = 8, N = \text{LCM} = 8$ |
| 80 Ba (b) 2  | $\sin(3\pi n/5) + \cos(4\pi n/7)$                 | $\frac{3}{10}$ and $\frac{2}{7} \rightarrow N_1 = 10, N_2 = 7, N = 70$           |
| 69 Bh (a) 4  | $\cos(11\pi n/3)$                                 | $\frac{11}{6} \rightarrow N = 6$                                                 |
| 69 Bh (b)    | $e^{j(7/5)n}$                                     | $\frac{7}{10\pi}$ , <b>irrational</b> $\rightarrow$ <b>aperiodic</b>             |
| 67 Mng (a) 4 | $\cos(5\pi n/3)$                                  | $\frac{5}{6} \rightarrow N = 6$                                                  |
| 67 Mng (b)   | $\sin(\pi n/\sqrt{2} + \pi/8)$                    | $\frac{1}{2\sqrt{2}}$ , <b>irrational</b> $\rightarrow$ <b>aperiodic</b>         |
| 66 Ma (a) 4  | $\cos(3\pi n/8 + \pi/4)$                          | $\frac{3}{16} \rightarrow N = 16$                                                |
| 66 Ma (b)    | $\sin(0.8n)$ : printed "sing(0.8n)"               | $\frac{0.4}{\pi}$ , <b>irrational</b> $\rightarrow$ <b>aperiodic</b>             |
| \$5 papers   | $\cos(2\pi n/5) + \sin(\pi n/3)$                  | $\frac{1}{5}, \frac{1}{6} \rightarrow N = \text{LCM}(5, 6) = 30$                 |
| \$5 papers   | $e^{j(\pi n/3 + \pi/4)}, e^{j(\pi n/2 + 4\pi/7)}$ | $\frac{1}{6} \rightarrow N = 6; \frac{1}{4} \rightarrow N = 4$                   |

**The rule in one line:** a  $\pi$  in  $\omega_0$  that cancels against the  $2\pi$  leaves a ratio of whole numbers and the signal is periodic. No  $\pi$ , or a  $\sqrt{\quad}$  left over, and it is not.

**7 Even and odd parts**

**HOT 4/45** 71 Ch 3, 70 Asa 3, 71 Shr 4+5, 67 Mng 2

**QUESTION AS PRINTED** 2071 Chaitra · Q1 · [3]

Find the even and odd part of the signal  $x[n] = 1$  for  $-4 \leq n \leq 0$ ;  $2$  for  $1 \leq n \leq 4$ .

**Worked, in the tabular form (use this layout every time)**

| $n$                                  | -4   | -3   | -2   | -1   | 0 | 1   | 2   | 3   | 4   |
|--------------------------------------|------|------|------|------|---|-----|-----|-----|-----|
| $x[n]$                               | 1    | 1    | 1    | 1    | 1 | 2   | 2   | 2   | 2   |
| $x[-n]$                              | 2    | 2    | 2    | 2    | 1 | 1   | 1   | 1   | 1   |
| $x_e[n] = \frac{1}{2}(x[n] + x[-n])$ | 1.5  | 1.5  | 1.5  | 1.5  | 1 | 1.5 | 1.5 | 1.5 | 1.5 |
| $x_o[n] = \frac{1}{2}(x[n] - x[-n])$ | -0.5 | -0.5 | -0.5 | -0.5 | 0 | 0.5 | 0.5 | 0.5 | 0.5 |

### Reading the table

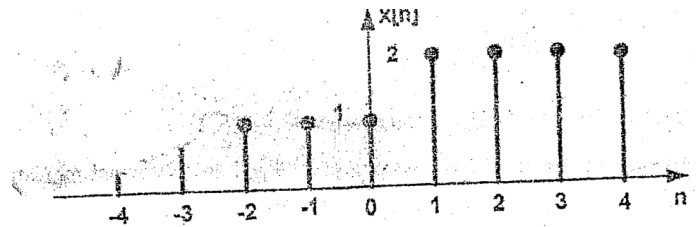
- Row 2 is row 1 **reversed about**  $n = 0$ :  $x[-4]$  takes the old  $x[4]$ , and so on.
- Checks that cost nothing**:  $x_e$  is symmetric about  $n = 0$  ✓, and  $x_o[0] = 0$  ✓
- Add the last two rows back together and row 1 must reappear ✓

$$x_e[n] = \{1.5, 1.5, 1.5, 1.5, \underline{1}, 1.5, 1.5, 1.5, 1.5\}$$

$$x_o[n] = \{-0.5, -0.5, -0.5, -0.5, \underline{0}, 0.5, 0.5, 0.5, 0.5\}$$

**67 Mng 2**:  $x(n) = 2u[n] - 2u[n-4]$

- That window is  $\{2, 2, 2, 2\}$  for  $n = 0 \dots 3$ , zero elsewhere.
- $x[-n] = \{2, 2, 2, 2\}$  for  $n = -3 \dots 0$ .
- $x_e = \{1, 1, 1, \underline{2}, 1, 1, 1\}$  and  $x_o = \{-1, -1, -1, \underline{0}, 1, 1, 1\}$  for  $n = -3 \dots 3$ .
- The  $n = 0$  sample doubles in  $x_e$** , because it overlaps itself.



71 Shr **4+5** gives the signal as a stem plot instead of a formula. Read the samples off the plot, then run exactly the same table. CT 704 2071 Shrawan Q1

### Second half of 71 Shr

- The same paper then gives  $x[n] = h[n] = 1$  for  $-1 \leq n \leq 1$ , zero elsewhere, and asks for the output by the graphical method.
- Both are  $\{1, 1, 1\}$  starting at  $n = -1$ , so the output starts at  $-2$  and has  $3 + 3 - 1 = 5$  samples.

$$y[n] = \{1, 2, \underline{3}, 2, 1\} \quad n = -2 \dots 2$$

- The triangle** is what convolving two equal rectangles always gives: worth recognising on sight.

## 8 Plotting and transformation of the independent variable

**HOT 4/45** 76 Ch **2+3**, 74 Ch **3+2**, 66 Ma **2**, 69 Bh **3** · **pure marks**: no formula to recall, only care with indices

**QUESTION AS PRINTED** 2076 Chaitra · Q1 · [2+3]

Define even and odd type discrete time signals with suitable example. Plot the signal  $x[-2n+3]$  where  $x[n] = \{1, 2, 0, -1, -3, -4\}$ .

**Method for  $x[an+b]$  (index map, never "shift then scale")**

- Take  $x[n] = \{1, 2, 0, -1, -3, -4\}$  starting at  $n = 0$ , so  $x[0] = 1, \dots, x[5] = -4$ .
- For each output index  $n$ , compute the argument  $m = -2n + 3$  and copy  $x[m]$ .
- Keep only those  $n$  for which  $m$  lands on an existing sample,  $0 \leq m \leq 5$ :

$$0 \leq -2n + 3 \leq 5 \Rightarrow -1 \leq n \leq 1.5 \Rightarrow n = -1, 0, 1$$

- Only three samples survive**, because scaling by 2 throws away every second one.

| $n$ | $m = -2n + 3$ | $x[m]$ | $y[n]$ |
|-----|---------------|--------|--------|
| -1  | 5             | -4     | -4     |
| 0   | 3             | -1     | -1     |
| 1   | 1             | 2      | 2      |

$$x[-2n+3] = \{-4, \underline{-1}, 2\} \quad n = -1, 0, 1$$

The even-indexed originals ( $x[0], x[2], x[4]$ ) are lost: that is decimation, and it is not reversible.

### The three plotting questions

- 74 Ch 3+2**:  $x[n] = u[n] - u[n-3] + 5\delta[n-4] + nu[n-6]$ 
  - $u[n] - u[n-3] = 1$  for  $n = 0, 1, 2$
  - $5\delta[n-4] = 5$  at  $n = 4$
  - $nu[n-6] = n$  for  $n \geq 6$ : 6, 7, 8, ... **a ramp that never stops**
  - $x[n] = \{1, 1, 1, 0, 5, 0, 6, 7, 8, \dots\}$
- 66 Ma 2**:
 
$$x(n) = u(n) - u(n-5) + 5\delta(n-6) + nu(n-7) - nu(n-9)$$
  - 1 for  $n = 0 \dots 4$ ; 5 at  $n = 6$ ; the two ramps leave only  $n = 7$  and  $n = 8 \rightarrow$  values 7 and 8.
  - $x[n] = \{1, 1, 1, 1, 1, 0, 5, 7, 8\}$ , zero after.
  - $nu(n-7) - nu(n-9)$  is a **windowed ramp**, not an infinite one: the second term cancels everything from  $n = 9$  on.
- 69 Bh 3**:  $x[n] = u[n+8] - u[n-4]$ 
  - A rectangle of height 1 from  $n = -8$  to  $n = 3$ : **12 samples**.
  - The upper limit is exclusive**:  $u[n-4]$  switches on at  $n = 4$ , so  $n = 3$  is the last non-zero sample.

## 9 Testing a system for its properties

**TOP 12/45** 76 Ch **5**, 74 Ch **3+3**, 82 Ba **2+2**, 78 Ch **3**, 75 Ash **3+3**, 71 Bh **4**, 70 Bh **5**, 68 Bh **5**, 67 Mng, 66 Ma **10**, 69 Bh **5**, 80 Bh **2+2+2**, 81 Ba **5**

**QUESTION AS PRINTED** 2068 Bhadra · Q1 · [5]

State whether or not the system  $y[n] = e^{x[2n]}$  is (a) linear (b) time invariant (c) memoryless (d) causal, where  $x[n]$  is input to system and  $y[n]$  is output of system.

**Worked in full:**  $y[n] = e^{x[2n]}$

- (a) Linear?
  - $x_1 \rightarrow e^{x_1[2n]}$ ,  $x_2 \rightarrow e^{x_2[2n]}$ .
  - Input  $ax_1 + bx_2$  gives  $e^{ax_1[2n] + bx_2[2n]} = e^{ax_1[2n]} \cdot e^{bx_2[2n]}$ .
  - Required:  $a e^{x_1[2n]} + b e^{x_2[2n]}$ . A product is not a weighted sum.
  - **Not linear** ×
- (b) Time invariant?
  - Delayed input:  $x_1[n] = x[n - n_0]$  gives  $y_1[n] = e^{x[2n - n_0]}$ .
  - Delayed output:  $y[n - n_0] = e^{x[2(n - n_0)]} = e^{x[2n - 2n_0]}$ .
  - $2n - n_0 \neq 2n - 2n_0$ , so **not time invariant** ×
  - **This is the general rule:** any time-scaled index  $x[2n]$ ,  $x[n^2]$ ,  $x[-n]$  makes a system time varying.
- (c) Memoryless?
  - At  $n = 3$  the output needs  $x[6]$ , a different instant.
  - **Not memoryless** × (it is dynamic).
- (d) Causal?
  - For  $n > 0$ ,  $2n > n$ : the output at  $n = 3$  needs  $x[6]$ , a future input.
  - **Not causal** ×
- **Stable?** (not asked here, but 66 Ma asks it)
  - $|x[n]| \leq M$  gives  $|y[n]| \leq e^M$ , finite  $\rightarrow$  **BIBO stable** ✓

**Answer shape that scores:** one line of algebra per property, then the verdict with its reason. "Not linear" alone gets nothing.

### Every other system the papers set, with the working reduced to its deciding line

The instruction, as the papers print it: "State whether or not the system ...is (a) linear (b) time invariant (c) memoryless (d) causal (e) stable", or "Determine whether the following systems are (a) causal (b) linear", or "Check whether the following system is linear or non-linear, time variant or invariant". **Name every property the paper lists and no more:** the marks are split one per property, so a full proof of two of five is worth two marks.

| Paper                  | System                                                   | Verdict, and the line that decides it                                                                                                               |
|------------------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 76 Ch <b>5</b>         | $y[n] = x[-n]$ (TI?)                                     | Delayed input $x[-n - n_0]$ ; delayed output $x[-(n - n_0)] = x[-n + n_0]$ . Different $\rightarrow$ <b>time varying</b>                            |
| 76 Ch <b>5</b>         | $y[n] = x[n^2]$ (linear?)                                | $ax_1[n^2] + bx_2[n^2]$ is exactly what superposition demands $\rightarrow$ <b>linear</b> ✓ (but time varying)                                      |
| 74 Ch, 78 Ch <b>3</b>  | $y[n] = y[n - 4] + x[n - 4]$ (TI?)                       | Every index is a plain delay, no explicit $n \rightarrow$ <b>time invariant</b> ✓                                                                   |
| 74 Ch, 75 Ash <b>3</b> | $y[n] = x^2[n]$ (linear?)                                | $(ax_1 + bx_2)^2$ leaves the cross term $2abx_1x_2 \rightarrow$ <b>not linear</b>                                                                   |
| 82 Ba <b>2+2</b>       | $y[n] = x[n] + x[-n]$ (causal? TI?)                      | At $n = -2$ it needs $x[2] \rightarrow$ <b>non-causal</b> ; the fold makes it <b>time varying</b>                                                   |
| 75 Ash <b>3</b>        | $y[n] = \cos(\frac{5\pi}{8}n + \frac{\pi}{4})$           | <b>No <math>x[n]</math> anywhere:</b> zero input gives non-zero output $\rightarrow$ <b>not linear</b>                                              |
| 71 Bh <b>4</b>         | $y[n] = x[n] - 3x[n - 1]$ ;<br>$y[n] = x[n + 1] + 4x[n]$ | Both <b>linear</b> ✓; first <b>causal</b> ✓, second needs $x[n + 1] \rightarrow$ <b>non-causal</b>                                                  |
| 70 Bh <b>5</b>         | $y(n) = \log_{10}[x(n)]$                                 | $\log(ax_1 + bx_2) \neq a \log x_1 + b \log x_2 \rightarrow$ <b>non-linear</b> ; TI ✓; causal ✓                                                     |
| 70 Bh <b>5</b>         | $y(n) = x(-n - 2)$                                       | <b>Linear</b> ✓; fold $\rightarrow$ <b>time varying</b> ; at $n = -3$ needs $x[1] \rightarrow$ <b>non-causal</b>                                    |
| 67 Mng                 | $y(n) = n x[2(n - 2)]$ ,<br>$n > 0$                      | <b>Linear</b> ✓; explicit $n$ and scaled index $\rightarrow$ <b>time varying</b> ; needs $x[2n - 4] \rightarrow$ <b>not memoryless</b>              |
| 66 Ma <b>10</b>        | $y(n) = 2^{\log_2 x(n)} + 2^{\log_2 x(n)}$               | This simplifies to $y(n) = 2x(n)$ : <b>linear</b> , TI, memoryless, causal, stable: <b>simplify before testing</b>                                  |
| 66 Ma <b>10</b>        | $y(n) = \sin\{x(n) - x(n - 1)\}$                         | <b>Non-linear</b> (sine of a sum); TI ✓; dynamic; causal ✓; <b>stable</b> ✓ since $ \sin  \leq 1$                                                   |
| 80 Bh <b>2+2+2</b>     | $y(n) = \sum_{k=0}^n x(k)$ (accumulator)                 | Needs all past inputs $\rightarrow$ <b>not memoryless</b> ; TI ✓; $x[n] = u[n]$ gives $y[n] = n + 1 \rightarrow \infty \rightarrow$ <b>unstable</b> |
| 81 Ba <b>5</b>         | $y(n) = x(n) + e^a y(n - 1)$                             | $h[n] = (e^a)^n u[n]$ , so $\sum  h $ converges only for $ e^a  < 1 \rightarrow$ <b>BIBO stable iff <math>a &lt; 0</math></b>                       |

## 10 Sampling an analog signal

**HOT 7/45** 81 Ch **1+3+3**, 74 Ma **7**, 78 Ch **2+2+3**, 80 Ch **3+4**, 73 Bh **3+4**, 73 Ma **4**, 76 Bh **2+1+1+2+2** · EX 753 sets this nearly every year

**QUESTION AS PRINTED** 2081 Chaitra · Q1 · [1+3+3]

Consider the analog signal  $x_a(t) = 3 \cos 100\pi t$ .

- Determine the minimum sampling rate required to avoid aliasing.
- Suppose that signal is sampled at  $F_s = 75$  Hz. What is the discrete time signal obtained after sampling?
- Assume now that we sample this signal using a sampling rate  $F_s = 5,000$  samples/sec. What is discrete time signal after sampling?

**Method (used for every problem in this section)**

- Read the analog frequency off the argument:  $\cos(2\pi Ft)$ , so  $2\pi F = 100\pi$  gives  $F = 50$  Hz. **Divide by  $2\pi$ , not by  $\pi$ .**
- Nyquist rate  $= 2F_{max}$ .
- Substitute  $t = nT = n/F_s$ : the digital frequency is  $\omega = \frac{2\pi F}{F_s}$  rad/sample.
- Fold  $\omega$  into  $-\pi \leq \omega < \pi$  by subtracting multiples of  $2\pi$ . If you had to subtract anything, **that term has aliased**: say so.
- Write the answer as  $x[n] = A \cos(\omega n)$  with the folded  $\omega$ .

**(a) Minimum sampling rate**

$$- F = \frac{100\pi}{2\pi} = 50 \text{ Hz.}$$

$$F_s \geq 2F = \boxed{100 \text{ Hz}}$$

**(b)  $F_s = 75$  Hz: below Nyquist**

- $\omega = \frac{2\pi \times 50}{75} = \frac{4\pi}{3}$  rad/sample.
- $\frac{4\pi}{3} > \pi$ , so fold:  $\frac{4\pi}{3} - 2\pi = -\frac{2\pi}{3}$ .
- Cosine is even, so

$$\boxed{x[n] = 3 \cos\left(\frac{2\pi n}{3}\right)}$$

- Aliased**: this is the same sequence a 25 Hz analog cosine would have given at 75 Hz. 75 Hz is below the 100 Hz Nyquist rate, exactly as part (a) warned.

**(c)  $F_s = 5000$  Hz: well above Nyquist**

$$- \omega = \frac{2\pi \times 50}{5000} = \frac{\pi}{50}, \text{ already inside } [-\pi, \pi).$$

$$\boxed{x[n] = 3 \cos\left(\frac{\pi n}{50}\right)}$$

no aliasing.

**Every other paper that runs this same method**

- 78 Ch 2+2+3**: same signal,  $F_s = 200$  Hz.  $\omega = \frac{2\pi \cdot 50}{200} = \frac{\pi}{2}$   
 $\rightarrow x[n] = 3 \cos(\pi n/2)$ , no aliasing.
- 80 Ch 3+4**:  
 $x(t) = \sin 2000\pi t - 5 \cos 12000\pi t + 10 \sin 6000\pi t$ ,  
 $F_s = 5000$ .  
 $- F = 1000, 6000, 3000$  Hz  $\rightarrow \omega = 0.4\pi, 2.4\pi, 1.2\pi$ .  
 $- \text{Fold: } 2.4\pi - 2\pi = 0.4\pi \text{ and } 1.2\pi - 2\pi = -0.8\pi$ .

$$x[n] = \sin(0.4\pi n) - 5 \cos(0.4\pi n) - 10 \sin(0.8\pi n)$$

- Two of the three aliased**, and the 6000 Hz term landed on top of the 1000 Hz term: they are now indistinguishable. The sign of the last term flips because sin is odd.
- For  $0 \leq n \leq 3$ :  $x[n] = \{-5, -6.472, 14.143, -6.053\}$  to 3 dp.
- 73 Bh 3+4**: identical but with  $+5 \cos 12000\pi t$ , so the middle term keeps its + sign.
- 73 Ma 4**:  
 $x(t) = \cos 1000\pi t + 3 \sin 3000\pi t + 5 \cos 5000\pi t$ ,  
 $F_s = 6000$ .  
 $- F = 500, 1500, 2500 \rightarrow \omega = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}$ : **all already inside  $[-\pi, \pi)$ , no aliasing**, because  
 $F_s = 6000 > 2 \times 2500$ .
- 76 Bh 2+1+1+2+2**:  
 $x(t) = 3 \cos 50\pi t + 10 \sin 300\pi t - 0.7 \cos 100\pi t$ ,  
 $F_s = 5000$ .  
 $- F = 25, 150, 50 \rightarrow F_{max} = 150$ , **Nyquist = 300 Hz**.  
 $- \omega = \frac{\pi}{100}, \frac{3\pi}{50}, \frac{\pi}{50}$ , **no aliasing**.  
 $- \text{Reconstruction: since } F_s \text{ is above Nyquist the original } x(t) \text{ comes back exactly, } x_r(t) = x(t). \text{ That is the whole answer to part (c): the sampling theorem guarantees it.}$

## 11 Difference equation: zero-input and zero-state response

PIN 1/45 78 Bh 4 · the LCCDE theory is in Ch 1 §1.9; the transform route to the same answer is Ch 2

QUESTION AS PRINTED 2078 Bhadra · Q4 · [4]

Determine the zero-input response for a second order system given by:  $y[n] - 3y[n-1] - 4y[n-2] = x[n]$

### Method

1. **Zero input** means set  $x[n] = 0$ , leaving the homogeneous equation  $y[n] - 3y[n-1] - 4y[n-2] = 0$ .
2. Try  $y[n] = \lambda^n$  and divide through by  $\lambda^{n-2}$  to get the **characteristic equation**.
3. Solve for the roots.
4. The response is a weighted sum of  $\lambda_i^n$ ; fix the weights from the initial conditions.

### Step-by-step

- $\lambda^n - 3\lambda^{n-1} - 4\lambda^{n-2} = 0 \Rightarrow \lambda^2 - 3\lambda - 4 = 0$
- $(\lambda - 4)(\lambda + 1) = 0 \Rightarrow \lambda_1 = 4, \lambda_2 = -1$

$$y_{zi}[n] = C_1(4)^n + C_2(-1)^n$$

### Fixing the constants

- **The paper gives no initial conditions**, so the boxed form above is the full answer. If  $y[-1]$  and  $y[-2]$  are supplied, solve

$$\frac{C_1}{4} - C_2 = y[-1], \quad \frac{C_1}{16} + C_2 = y[-2]$$

$$C_1 = \frac{16(y[-1] + y[-2])}{5}, \quad C_2 = \frac{4y[-2] - y[-1]}{5}$$

verified/added

- **Read the roots:**  $|\lambda_1| = 4 > 1$ , so the natural response **grows without bound** and the system is unstable. Saying that earns the last mark.

### The other half, if asked

- **Zero-state** response: set the initial conditions to zero and convolve,  $y_{zs}[n] = x[n] * h[n]$ , or use  $H(z)$  (Ch 2).
- **Total** =  $y_{zi} + y_{zs}$ . An LTI system must be relaxed, so a question that gives non-zero initial conditions is not asking about an LTI system.

## Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full. → reads “leads to / therefore”, + reads “and”, = reads “is”. ✓ means the property holds, × that it does not.

| Short | Means                                           | Short | Means                                  |
|-------|-------------------------------------------------|-------|----------------------------------------|
| DSP   | digital signal processing                       | DSAP  | digital signal analysis and processing |
| DT    | discrete-time                                   | CT    | continuous-time                        |
| LTI   | linear time-invariant                           | LSI   | linear shift-invariant, the same thing |
| TI    | time-invariant                                  | BIBO  | bounded input, bounded output          |
| LCCDE | linear constant-coefficient difference equation | ROC   | region of convergence                  |
| DTFS  | discrete-time Fourier series                    | DTFT  | discrete-time Fourier transform        |
| CTFT  | continuous-time Fourier transform               | DFS   | discrete Fourier series                |
| DFT   | discrete Fourier transform                      | IDFT  | inverse DFT                            |
| FFT   | fast Fourier transform                          | DIT   | decimation in time                     |
| DIF   | decimation in frequency                         | MAC   | multiply and accumulate                |
| FIR   | finite impulse response                         | IIR   | infinite impulse response              |
| LPF   | low pass filter                                 | HPF   | high pass filter                       |
| BPF   | band pass filter                                | BSF   | band stop filter                       |
| BLT   | bilinear transformation                         | II    | impulse invariance                     |
| A/D   | analog to digital                               | D/A   | digital to analog                      |
| S/H   | sample and hold                                 | ZOH   | zero-order hold                        |
| SNR   | signal to noise ratio                           | PYQ   | past-year question                     |
| coeff | coefficient                                     | freq  | frequency                              |
| eqn   | equation                                        | no.   | number                                 |
| ops   | operations                                      | attr  | attribute                              |
| w/    | with                                            | w/o   | without                                |
| Eg    | for example                                     | Vs    | compare / differentiate                |
| +Fig  | a figure is wanted                              | (I)   | importance / need                      |
| (+)   | advantage                                       | (-)   | disadvantage                           |

## Exam paper codes

### Month codes

- **Ba** Baishakh **Jth** Jestha **Asa** Ashad
- **Shr** Shrawan **Bh** Bhadra **Ash** Ashwin
- **Ka** Kartik **Mng** Mangsir **Po** Poush
- **Ma** Magh **Ch** Chaitra
- **Ash and Asa are different months**: 2075 Ashwin and 2070 Ashad are separate papers.

### How a year code is written

- **Bold** = **Regular** exam. Plain = Back exam.
  - “New Back (2066 & Later Batch)” counts as Back.
  - A header printing “Regular / Back” in one box counts as Regular.
- Typewriter = the EX 753 paper (DSP, BEX, IV/II). Normal font = **CT 704** (DSAP, BEI/BCT, IV/I).
- The two axes are independent: **76 Ch** is CT 704 Regular, 76 Ash is CT 704 Back, 76 **Bh** is EX 753 Regular, 74 **Ma** is EX 753 Back.
- **69 Bh and 69 Bh** are two different papers, one in each course.

## Frequency chips

- **TOP n/45** asked in **10 or more** of the 45 papers.
- **HOT n/45** asked in **4 to 9** papers.
- **PIN n/45** asked in **1 to 3** papers.
- **verified/added** written or corrected against a reference because the source notes did not cover it, or covered it wrongly.
- **5** marks the question has actually carried. Several chips mean the same question has been set at different marks in different years.
- The count is over **45 papers**: 28 CT 704 (2066 Magh to 2082 Bhadra) and 17 EX 753 (2069 Bhadra to 2081 Chaitra).
- Counts come from the Detailed PYQ document, which reproduces every question in full.